



Software Design Specifications

Intelligent System for Pakistan Sign Language Learning and Translation

Version: 1.4

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Definition of Terms, Acronyms and Abbreviations

Term	Description
SDS	Software Design Specification
PSL	Pakistan Sign Language
UI/UX	User Interface / User Experience
AI/ML	Artificial Intelligent / Machine Learning
GB	Gigabyte
RAM	Random-access Memory
API	Application Programming Interface
CRUD	Create, Read, Update, and Delete
ERD	Entity Relationship Diagram
PK	Primary Key
DB	Database

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1 Introduction

1.1 Purpose of Document

The purpose of this Software Design Specification (SDS) document is to describe the architectural, software, and design aspects of the Pakistan Sign Language (PSL) Translation and Learning Application. It defines how the system's functional and non-functional requirements will be implemented through appropriate software design and architecture.

The SDS serves the following objectives:

- To provide a structured design reference for all stakeholders, including developers, testers, and supervisors, and jury.
- To ensure that the design aligns with the system's objectives of accessibility, accuracy, and scalability.
- To provide a basis for implementation, integration, and verification activities.
- To support requirement traceability throughout the project lifecycle and maintain consistency between design and requirements.

1.2 Intended Audience

This Software Design Specification (SDS) is intended for the following stakeholders:

- **Project Sponsors and Management:** To understand the project objectives, scope, and resource requirements.
- **Developers and Engineers:** To obtain detailed functional and non-functional requirements necessary for design and implementation.
- **Testers and Quality Assurance Team:** To use the defined requirements as the basis for preparing test cases and validating system functionality.
- **UI/UX Designers:** To align interface designs with the requirements for usability and accessibility.
- **Documentation Writers:** To prepare user manuals, help guides, and training materials.
- **End Users (Non-hearing community representatives, educators, and learners):** To validate that the requirements address real-world needs and usability expectations.

1.3 Document Convention

- Font Family = Arial
- Font Size = 16 for headings, 14 for sub-headings, 10 for the rest of the content

1.4 Project Overview

The PSL Translation and Learning System is a mobile-based AI-driven application designed to bridge communication gaps between hearing and non-hearing individuals in Pakistan. The system uses Computer Vision, Machine Learning, and Natural Language Processing (NLP) to translate sign gestures to text and vice versa.

It also includes a learning module, where users can practice PSL through structured lessons and quizzes, supported by a dictionary module for reference. The backend integrates trained AI models with a cloud database, while the frontend (Flutter) provides an interactive user experience.

1.5 Scope

Included Functionalities:

- PSL-to-Text Translation using AI-based gesture recognition models.
- PSL Dictionary Module organized alphabetically and by category.
- Learning Module with structured lessons, animations, and practice tasks.
- Assessment Module for evaluating user performance using AI grading.
- User Authentication and Profile Management (login, signup, progress tracking).
- Gamification Elements (streaks, badges).

Excluded Functionalities:

- Support for other sign languages (e.g., ASL, BSL).
- Offline AI inference (requires internet connectivity for most operations).
- Advanced 3D avatar rendering.
- Integration with external social media or payment systems.

2 Design Considerations

2.1 Assumptions and Dependencies

The development and operation of the “Intelligent System for Pakistan Sign Language (PSL) Learning and Translation” relies on the following assumptions and dependencies:

- **Assumptions**
 - Users will have access to Android devices running Android 9.0 or higher with sufficient hardware resources (≥ 4 GB RAM, camera, and microphone).
 - A stable internet connection will be available for real-time translation, cloud storage, and updates.
 - PSL datasets can be collected, augmented, and used for training AI models without major legal or licensing restrictions.
 - Users are expected to have basic smartphone literacy to interact with the app.
- **Dependencies**
 - Dependence on third-party frameworks and libraries such as TensorFlow/PyTorch, OpenCV, and MediaPipe for AI and computer vision tasks.
 - Dependence on Firebase for data storage, user authentication, and synchronization.
 - Dependence on cloud infrastructure for backend services and model deployment.
 - Dependence on external APIs for certain translation features.
 - Future system upgrades depend on the availability of additional PSL datasets and feedback from the Deaf community.

2.2 Risks and Volatile Areas

- **Dataset Limitations:** Availability and quality of PSL datasets may affect model accuracy.
- **Model Performance:** Variations in lighting, signer style, or background could impact recognition.
- **Technology Upgrades:** Frameworks like TensorFlow and Flutter may change rapidly, requiring periodic updates.
- **User Diversity:** Differences in dialects or regional variations of PSL could affect translation consistency.
- **System Scalability:** Cloud infrastructure must handle increased user load if the system is deployed publicly.

To mitigate these risks, the system will be designed using a modular architecture, allowing updates to AI models, datasets, and UI independently. Version control and regular validation testing will ensure adaptability.

3 System Architecture

3.1 System Level Architecture

The system is divided into four main components:

- Mobile Application (Frontend) – User interface developed in Flutter.
- Application Server (Backend) – Python-based API using Flask/FastAPI for handling requests.
- AI Models – Deep learning models for gesture recognition and translation.
- Database – Firebase for storing user data, lessons, and dictionary content.

3.2 Software Architecture

The system follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer.

- User Interface Layer: Provides users with access to translation, learning, and assessment features.
- Middle Tier: Handles API communication, model inference, authentication, and business logic.
- Data Access Layer: Manages CRUD operations for user profiles, lessons, and translations.
- AI/ML Engine: Performs real-time PSL recognition and translation.

4 Design Strategy

The design emphasizes modularity, scalability, and user accessibility.

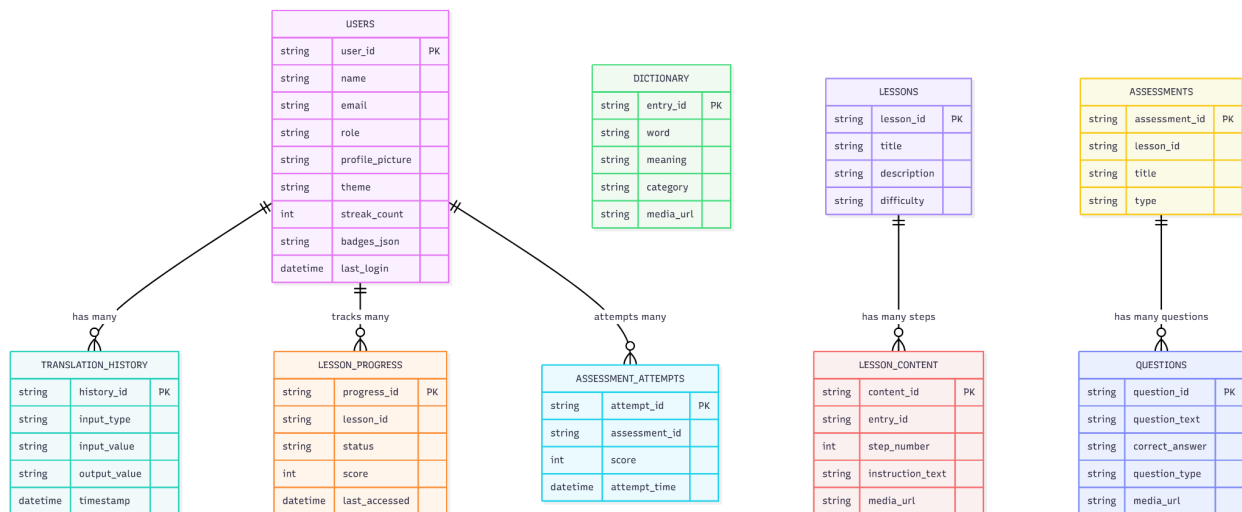
- **Future System Extension:** Architecture allows new sign languages and additional learning modules to be added without major redesign.
- **System Reuse:** AI components (gesture recognition and text-to-sign translation) can be reused in other accessibility applications.
- **User Interface Paradigm:** Intuitive, mobile-friendly UI for accessibility.
- **Data Management:** Firebase ensures flexible, document-based storage for sign records and user data.
- **Concurrency & Synchronization:** The backend handles concurrent API calls through asynchronous processing in FastAPI, ensuring smooth performance during real-time translations.

Trade-offs include prioritizing cloud-based model inference (for accuracy and scalability) over on-device inference (for speed and offline use).

5 Detailed System Design

5.1 Database Design

5.1.1 ER Diagram



5.1.2 Data Dictionary

5.1.2.1 Data 1

Users						
Name	Users					
Alias	App User / Learner					
Where-used/how-used	Stores all users who log in to the PSL application, including their personal information, progress, and settings. Used for authentication, profile display, storing progress, and personalization.					
Content description	Contains identifying information and user-level profile data.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
user_id	Unique system-generated identifier	String		NO	Auto	PK
name	Full name of the user	String		NO	None	
email	Unique email for login	String		NO	None	Unique
profile_picture	URL to stored image	String		YES	NULL	
streak_count	Number of consecutive days active	Int		YES	0	
badges	Earned badges stored as JSON	Map		YES	{}	
last_login	Time of last login	Timestamp		YES	NULL	

5.1.2.2 Data 2

Dictionary						
Name	Dictionary					
Alias	PSL Lexicon					
Where-used/how-used	Stores PSL words with meanings and media files (videos). Used in dictionary module and lessons					
Content description	A collection of PSL vocabulary entries.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
entry_id	Unique identifier	String		NO	Auto	PK

word	PSL Word	String		NO	None	
meaning	English Meaning	String		NO	None	
category	Category like colors, verbs, fruits	String		YES	None	
media_url	URL of the gesture video/image	String		YES	NULL	

5.1.2.3 Data 3

Lessons						
Name	Lessons					
Alias	Learning Module					
Where-used/how-used	Used in training module to deliver structured PSL lessons					
Content description	Metadata about each lesson.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
lesson_id	Unique lesson ID	String		NO	Auto	PK
title	Lesson title	String		NO	None	
description	Summary of lesson contents	String		YES	None	
difficulty	Level (Beginner/Intermediate/Advanced)	String		YES	"Beginner"	

5.1.2.4 Data 4

Lessons Content						
Name	Lesson Content					
Alias	Lesson Steps					
Where-used/how-used	Represents the individual steps or words displayed inside a lesson.					
Content description	Contains step-by-step learning segments.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
content_id	Unique Step ID	String		NO	Auto	PK

lesson_id	ID of parent lesson	String		NO	-	Reference
entry_id	ID of dictionary word used in step	String		YES	None	Reference
step_number	Order of the step	Int		NO	None	
instruction text	Text instruction	String		YES	None	
media_url	Video URL	String		YES	None	

5.1.2.5 Data 5

Assessments						
Name	Assessments					
Alias	Tests					
Where-used/how-used	Used to evaluate users after lessons					
Content description	Stores metadata for each assessment.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
assessment_id	Unique assessment ID	String		NO	Auto	PK
lesson_id	Linked lesson	String		YES	None	Reference
title	Title of quiz/test	String		No	None	
type	MCQ / gesture recognition	String		YES	None	

5.1.2.6 Data 6

Questions						
Name	Questions					
Alias	Assessment Questions					
Where-used/how-used	Used under assessments to test user knowledge.					
Content description	A pool of assessment questions.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
question_id	Unique question ID	String		NO	Auto	PK

question_text	Question statement	String		NO	None	
current_answer	Expected answer	String		NO	None	
question_type	MCQ / open-ended / gesture	String		YES	None	
media_url	Supporting media	String		YES	None	

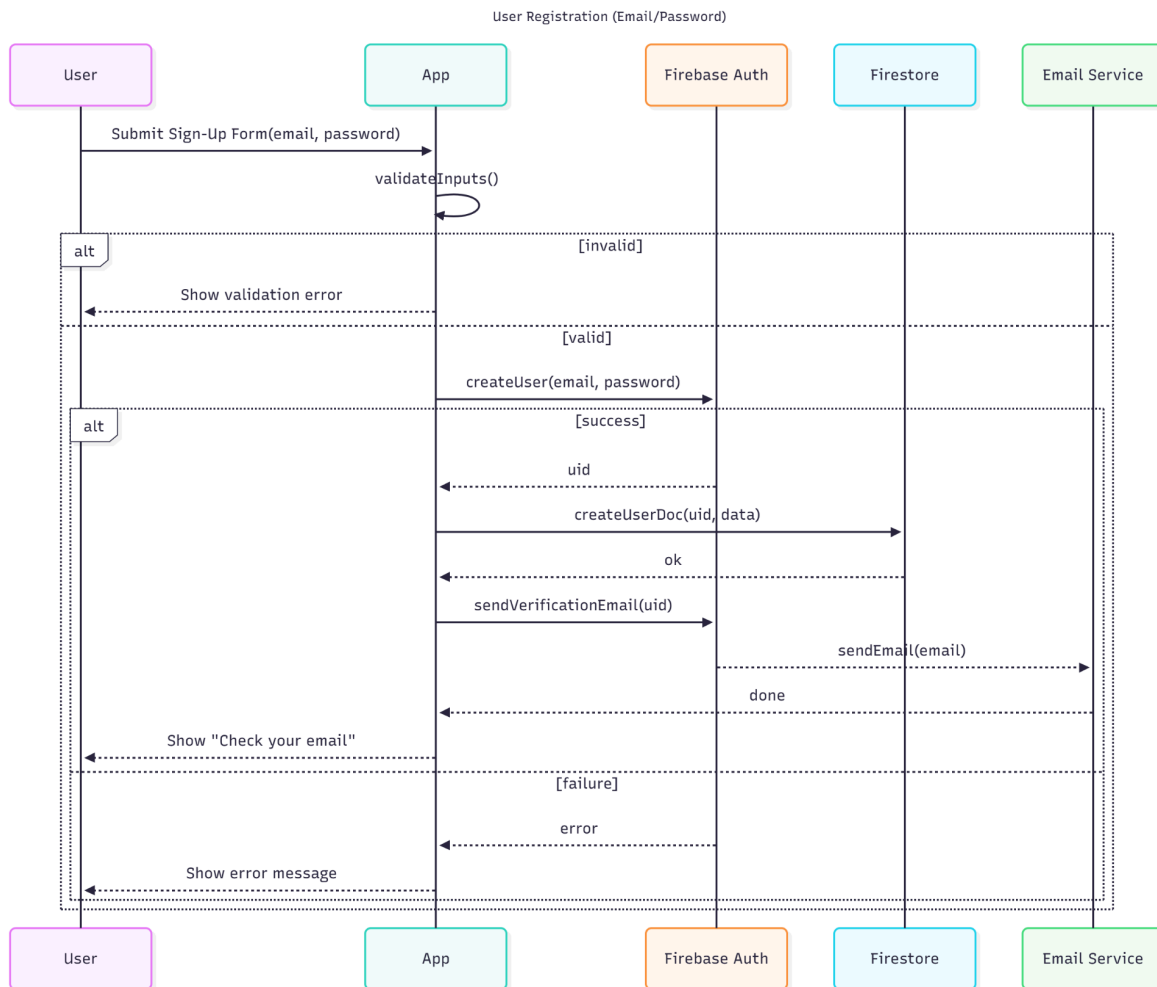
5.1.2.7 Data 7

Assessment Attempts (User Subcollection)						
Name	Assessment Attempts					
Alias	User Test Performances					
Where-used/how-used	Tracks how a user performed in assessments.					
Content description	Individual results for each attempt.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
attempt_id	Unique result ID	String		NO	Auto	PK
assessment_id	Assessment attempted	String		NO	None	Reference
user_id	User who attempted	String		NO	None	Reference
score	Score achieved	Int		YES	None	
attempt_time	When attempt was recorded	Timestamp		YES	Auto	

5.2 Application Design

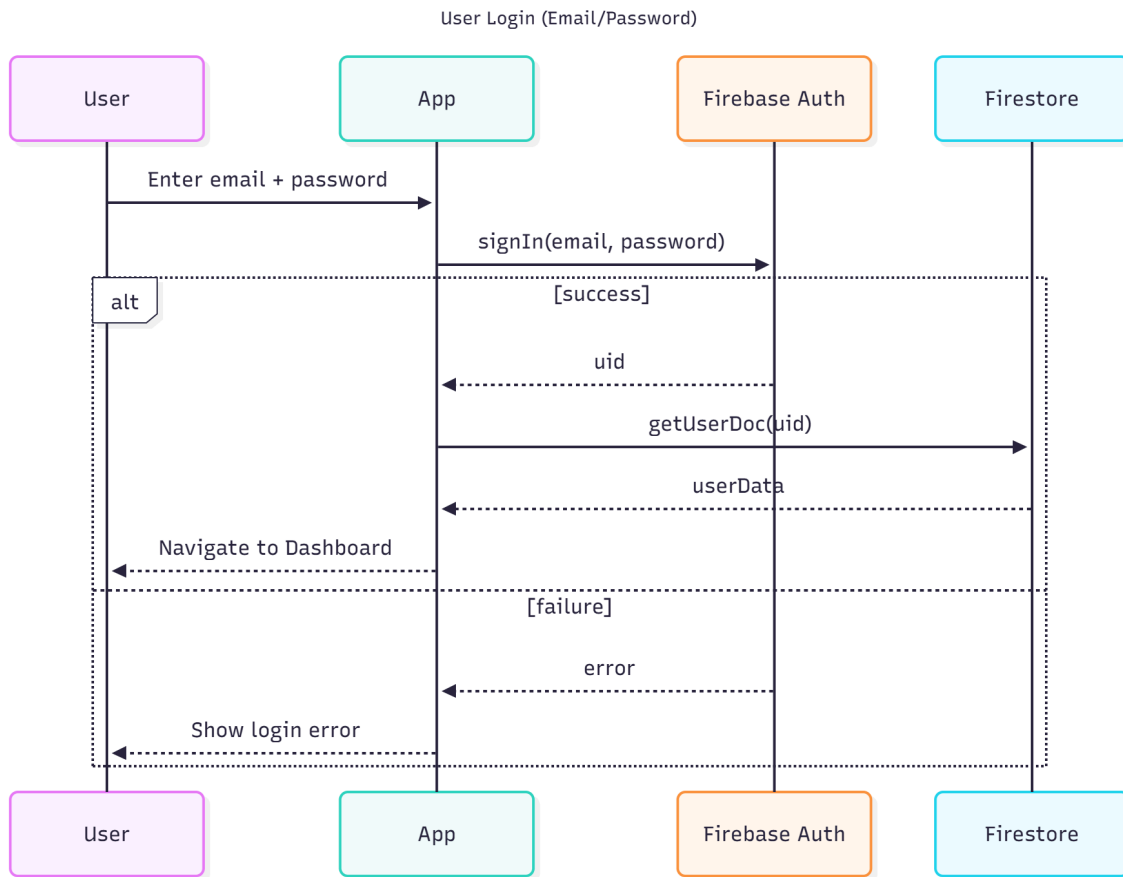
5.2.1 Sequence Diagram

5.2.1.1 Sequence Diagram 1



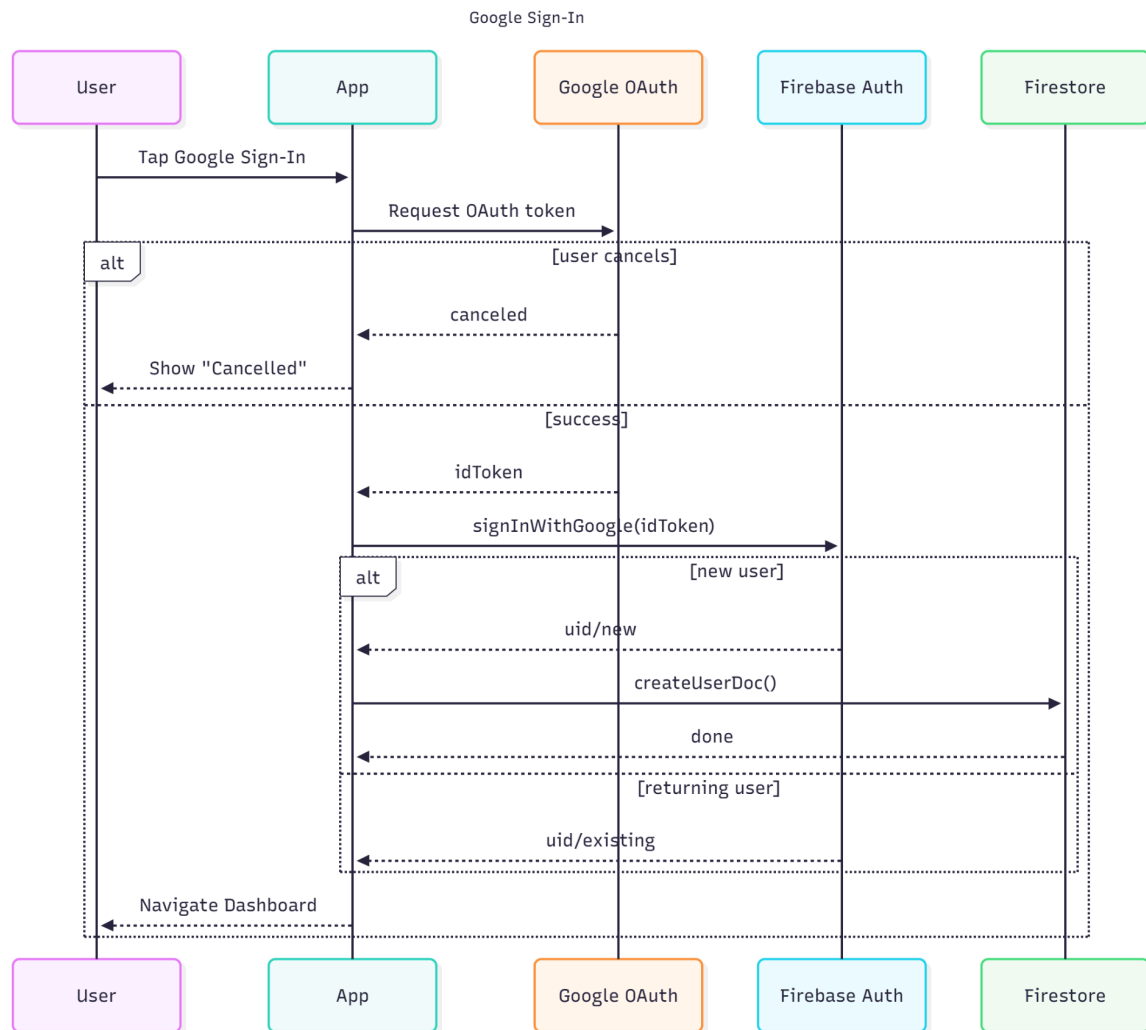
- User opens the app and selects Sign Up.
- App validates input fields (email, password).
- Backend checks whether the email already exists.
- If email is NOT unique → backend returns error → app shows error message.
- If email is unique → backend creates user record.
- Firebase/Auth service (or backend API) sends verification email.
- System responds with "Signup Successful - please verify."
- User is redirected to login screen.

5.2.1.2 Sequence Diagram 2



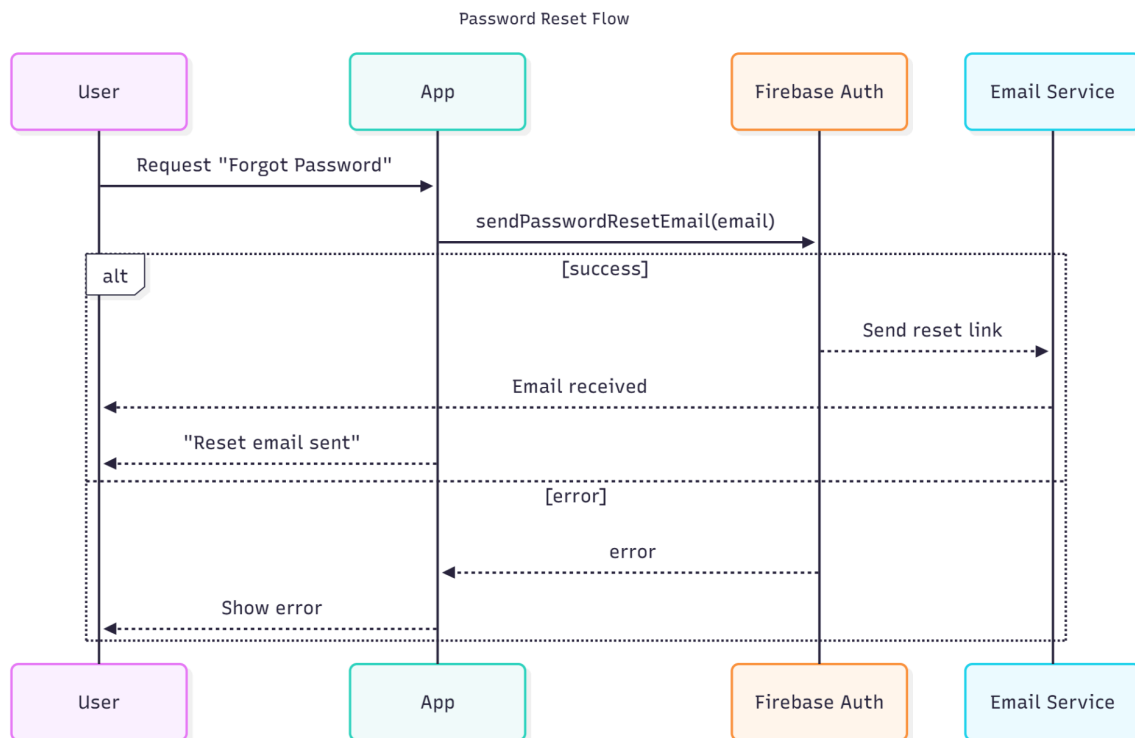
- User opens the Login screen.
- User enters email + password and submits.
- App sends credentials to backend.
- Backend verifies account existence.
- Password is checked.
- If incorrect → error returned → UI shows “Invalid Credentials.”
- If correct → system creates a valid session/token.
- User is redirected to Dashboard screen.

5.2.1.3 Sequence Diagram 3

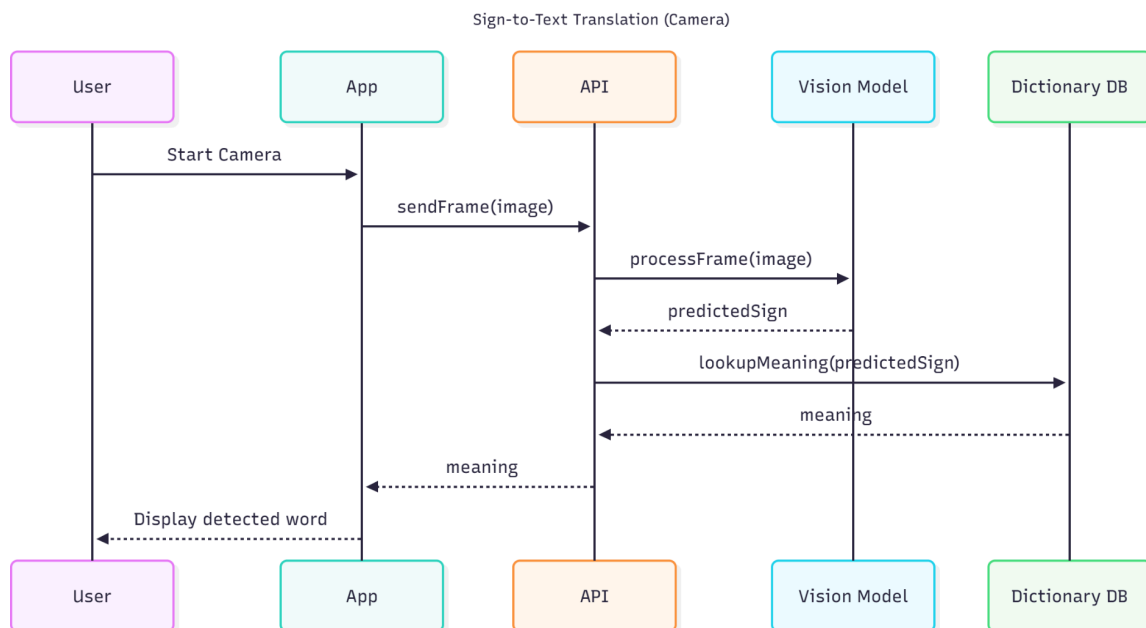


- User signs in using Google OAuth.

5.2.1.4 Sequence Diagram 4



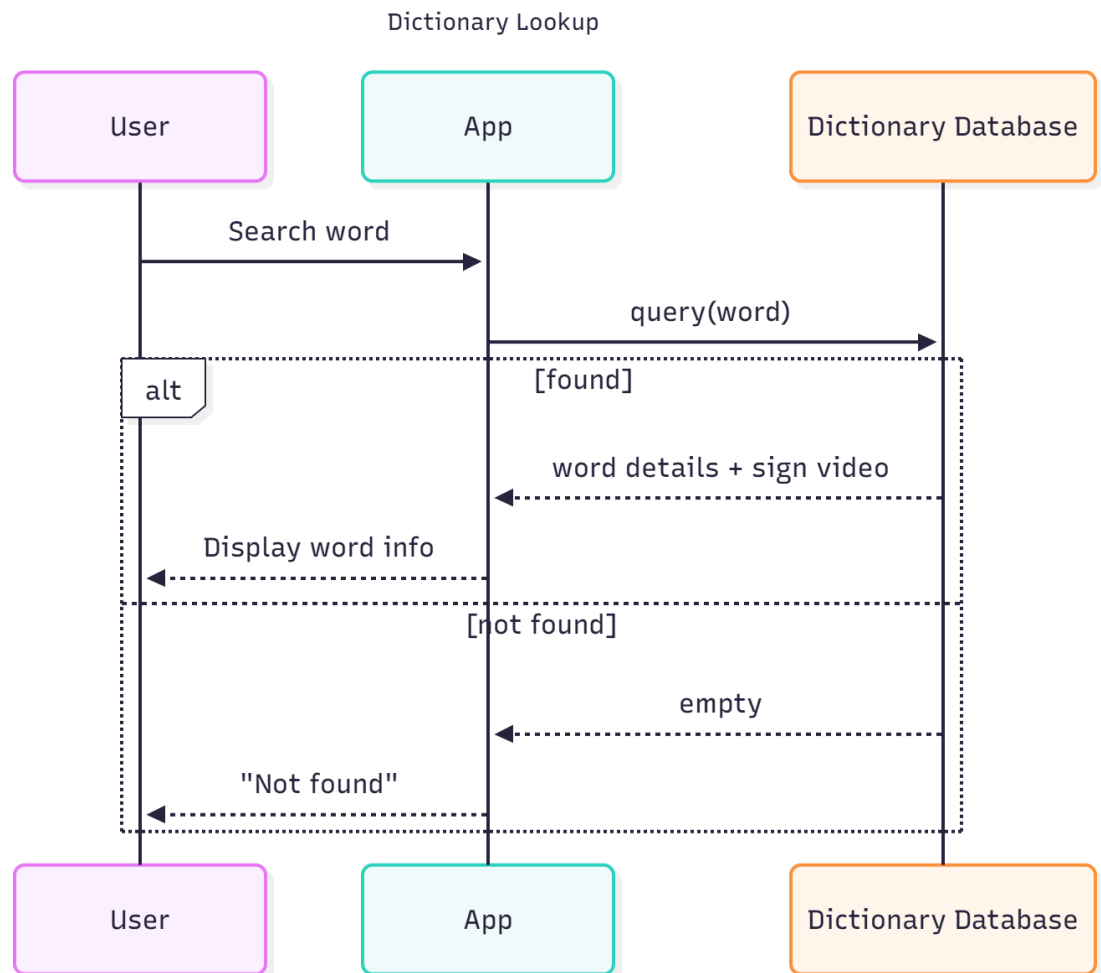
5.2.1.5 Sequence Diagram 5



- User activates camera translation.
- Camera stream is sent frame-by-frame to backend model.

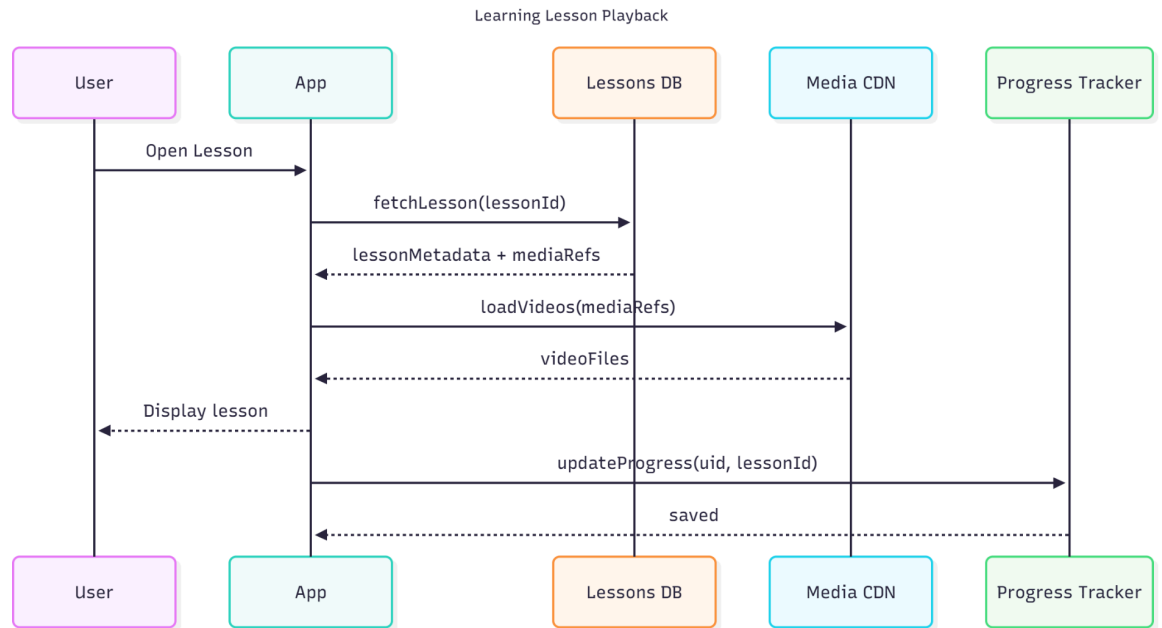
- Model performs hand detection + landmark extraction.
- Model classifies gesture into PSL label.
- Classification result is returned to app.
- App displays recognized text live.
- If recognition confidence is low → app shows “Uncertain Gesture.”

5.2.1.6 Sequence Diagram 6



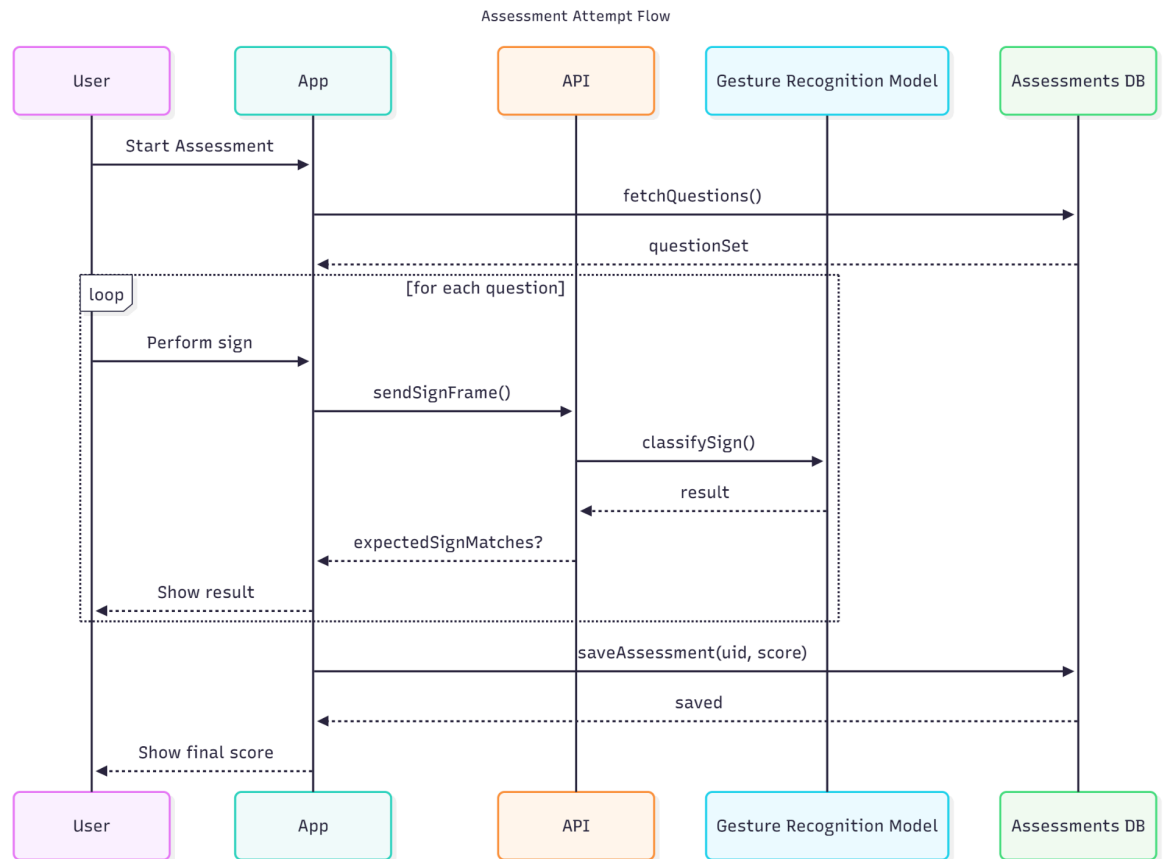
- User opens Dictionary module.
- User types a search term.
- App sends search query to backend/database.
- Backend checks for sign-word match.
- If no match → return “Not Found” to UI.
- If match exists → backend fetches sign video / image sequence.
- UI displays the sign representation, definition, and related words.
- User may repeat search or navigate back.

5.2.1.7 Sequence Diagram 7



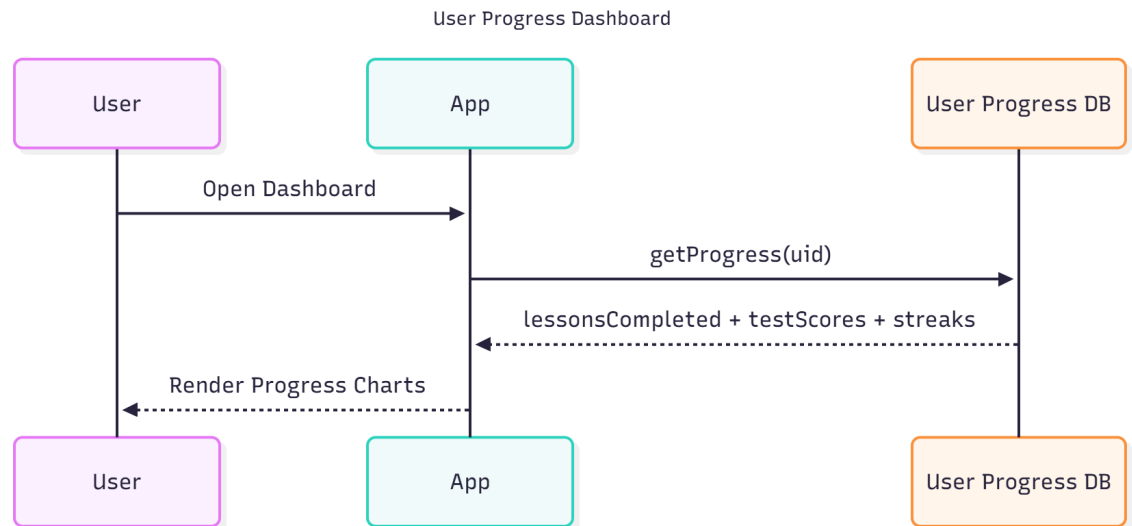
- User opens the Learning Module.
- User selects a lesson/chapter.
- App requests lesson content from backend.
- Backend returns lesson metadata + instructional videos.
- UI loads lesson structure.
- User watches videos, follows steps, and completes lesson.
- System records progress in user profile.
- User can proceed to next lesson or exit.

5.2.1.8 Sequence Diagram 8

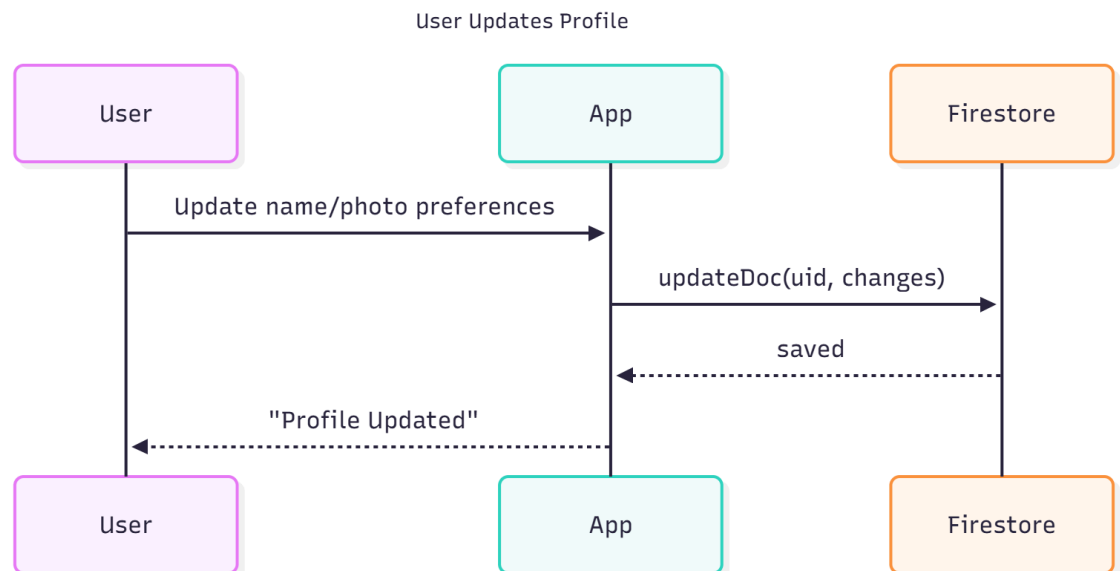


- User selects an assessment test.
- App requests quiz data & gesture tasks from backend.
- System loads questions and evaluation criteria.
- User answers MCQs or performs signs.
- For gesture tasks → camera captures signs and sends frames.
- Backend model evaluates sign accuracy.
- App records correctness for each question.
- At completion → backend calculates score and saves it.
- User receives results + unlocked achievements.

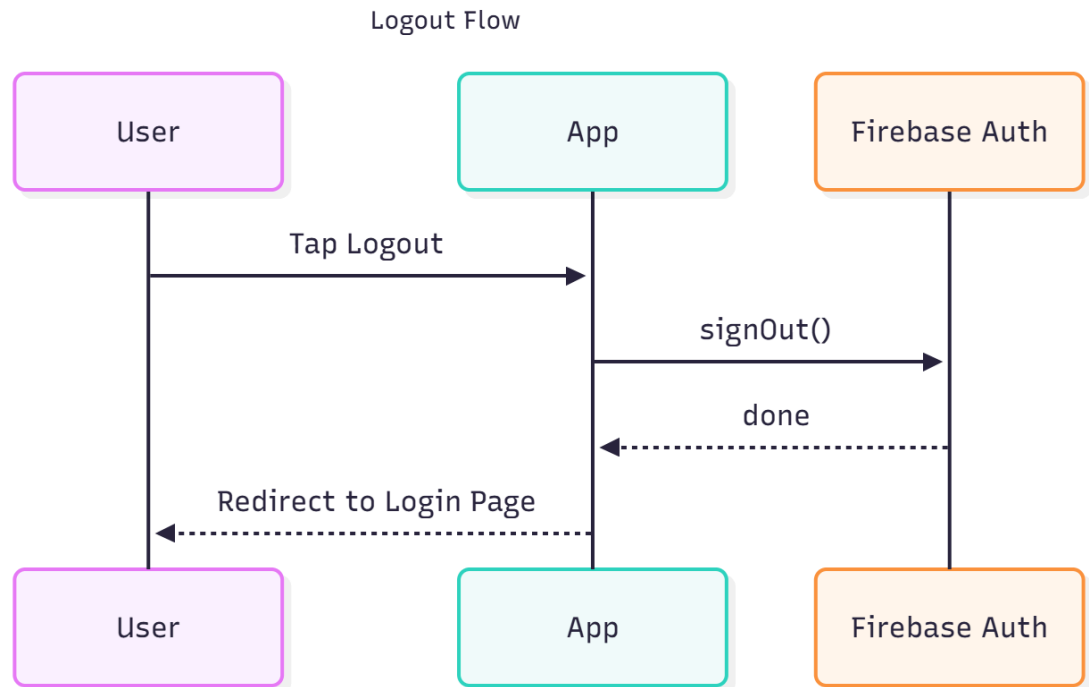
5.2.1.9 Sequence Diagram 9



5.2.1.10 Sequence Diagram 10

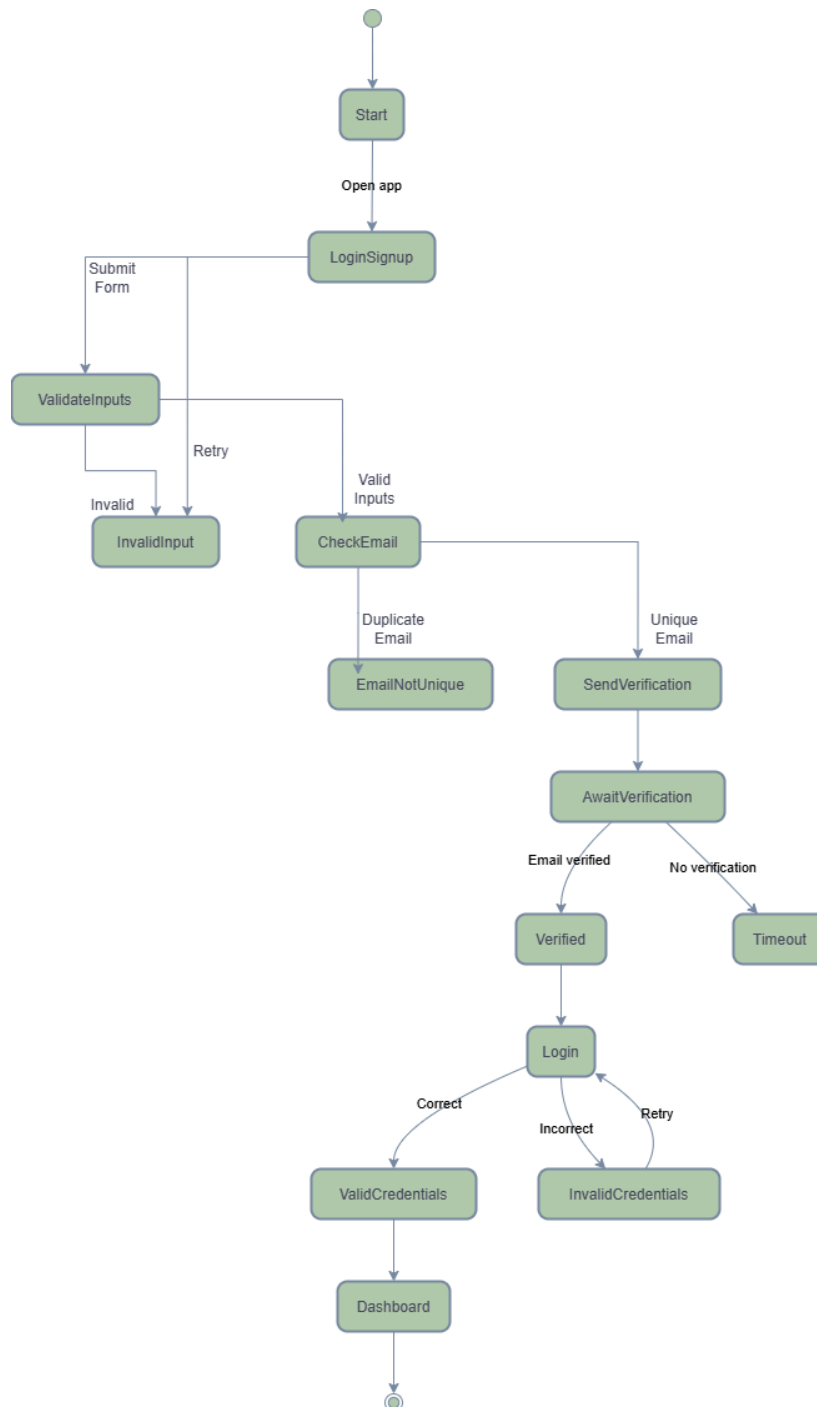


5.2.1.11 Sequence Diagram 11



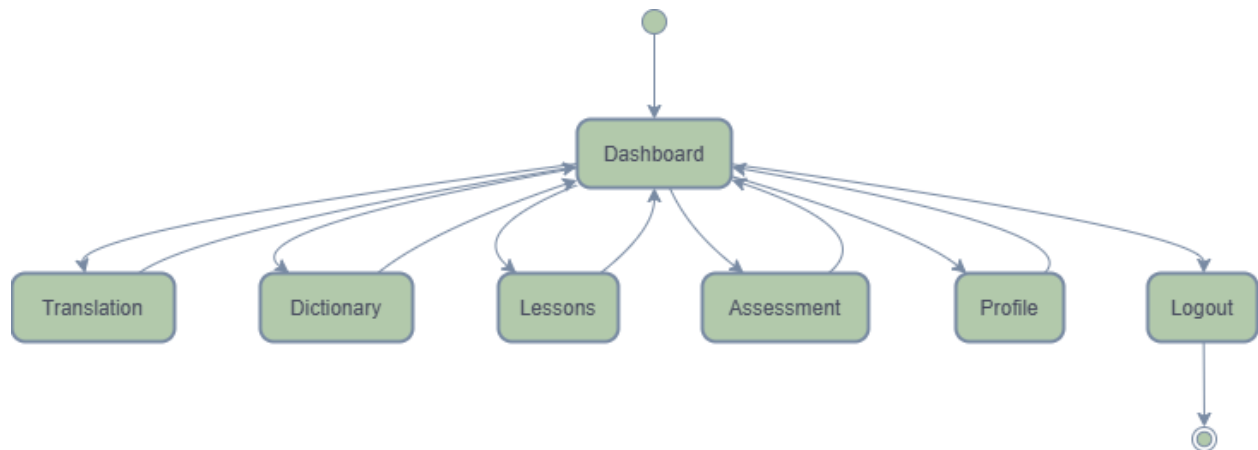
5.2.2 State Diagram

5.2.2.1 State Diagram 1



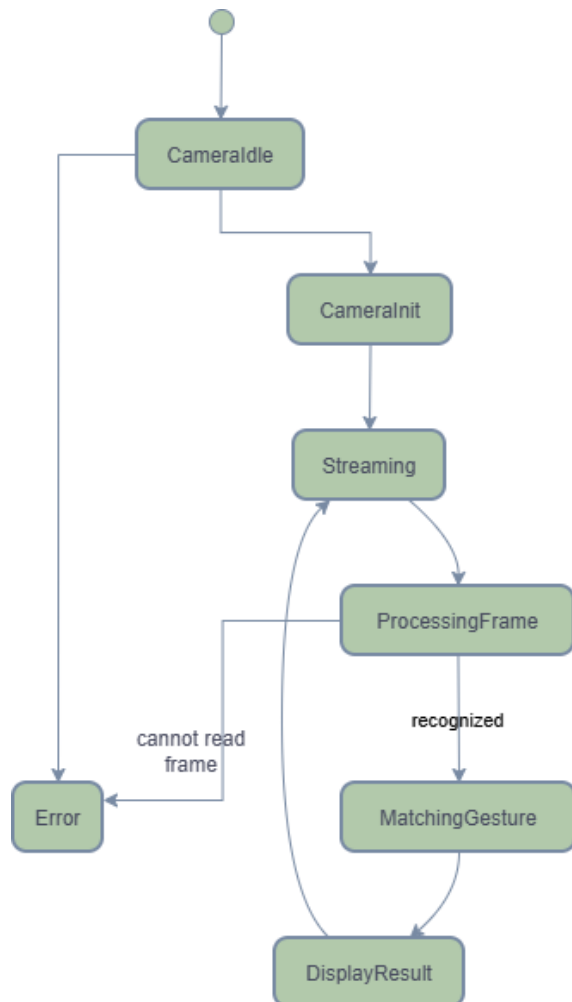
Represents how a user transitions through different authentication states: signup, login, verification, and error handling.

5.2.2.2 State Diagram 2



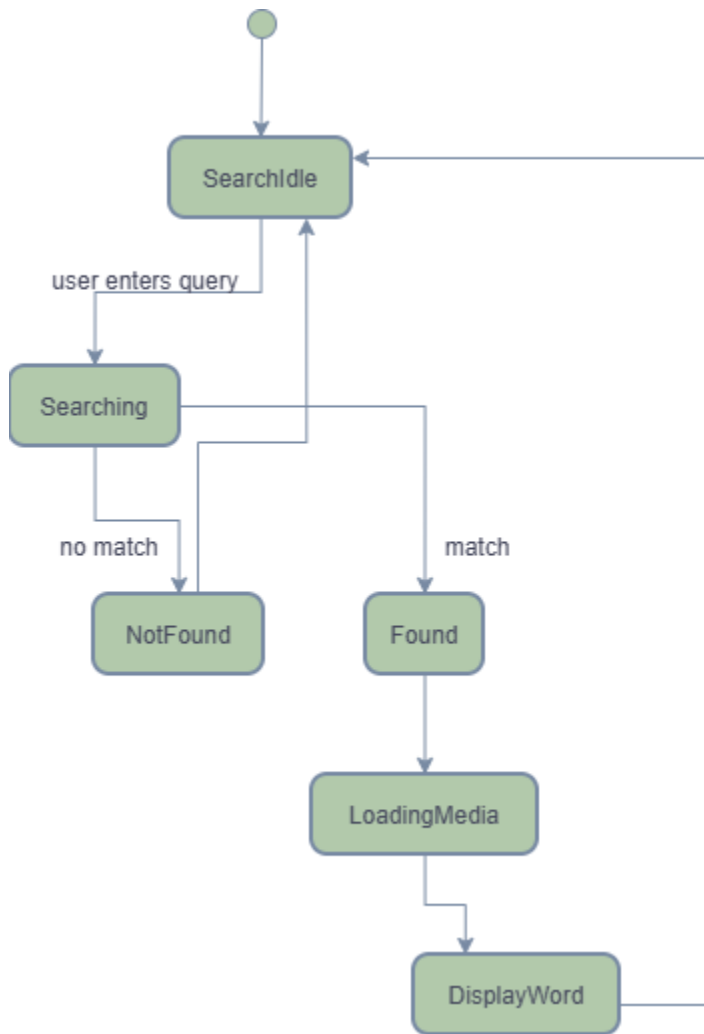
After login, the user can browse lessons, translate signs, access the dictionary, or attempt assessments.

5.2.2.3 State Diagram 3



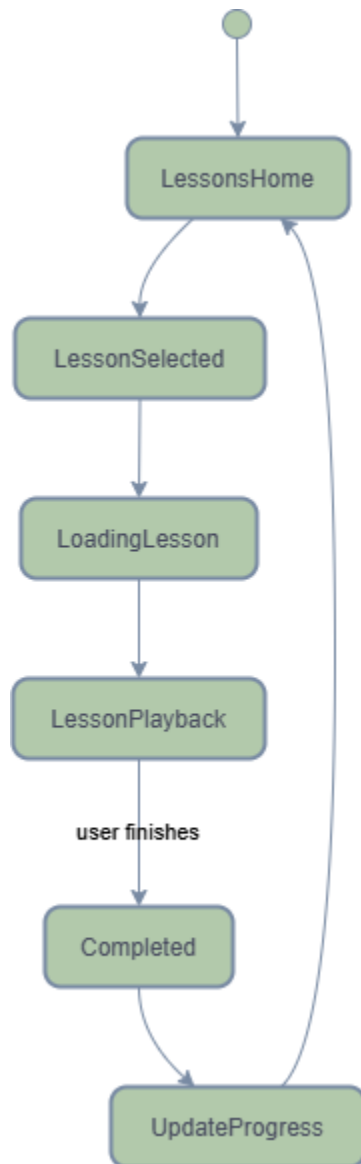
User uses device camera to translate hand gestures into text.

5.2.2.4 State Diagram 4



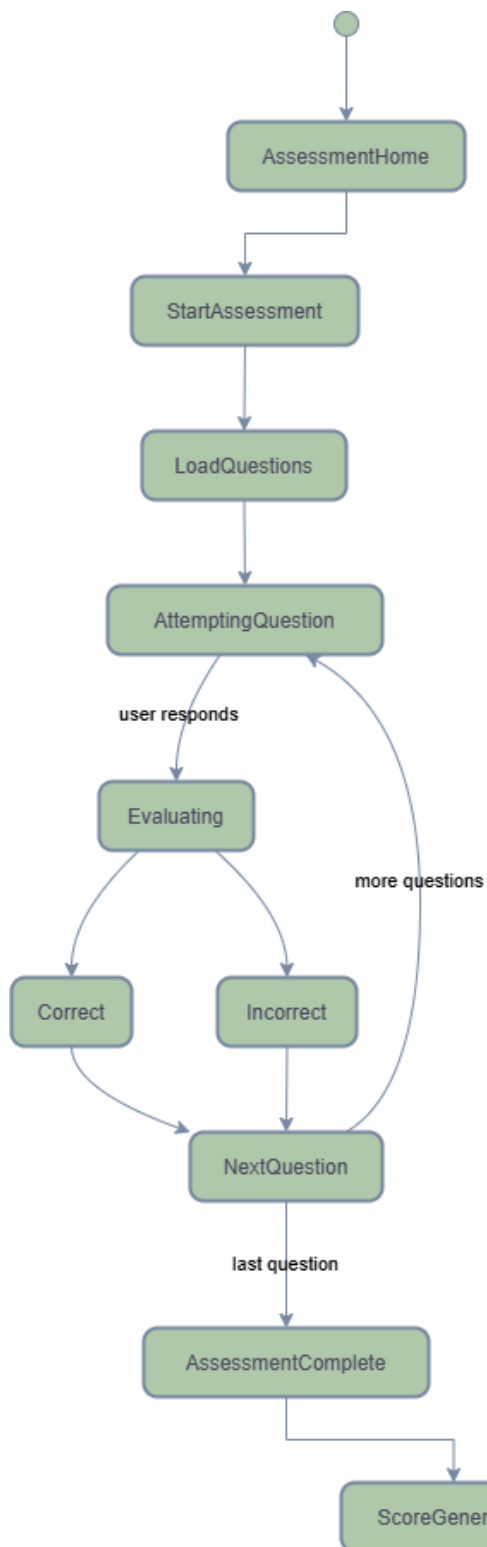
Handles word searching, selection, media loading, and fallback when a word is not found.

5.2.2.5 State Diagram 5



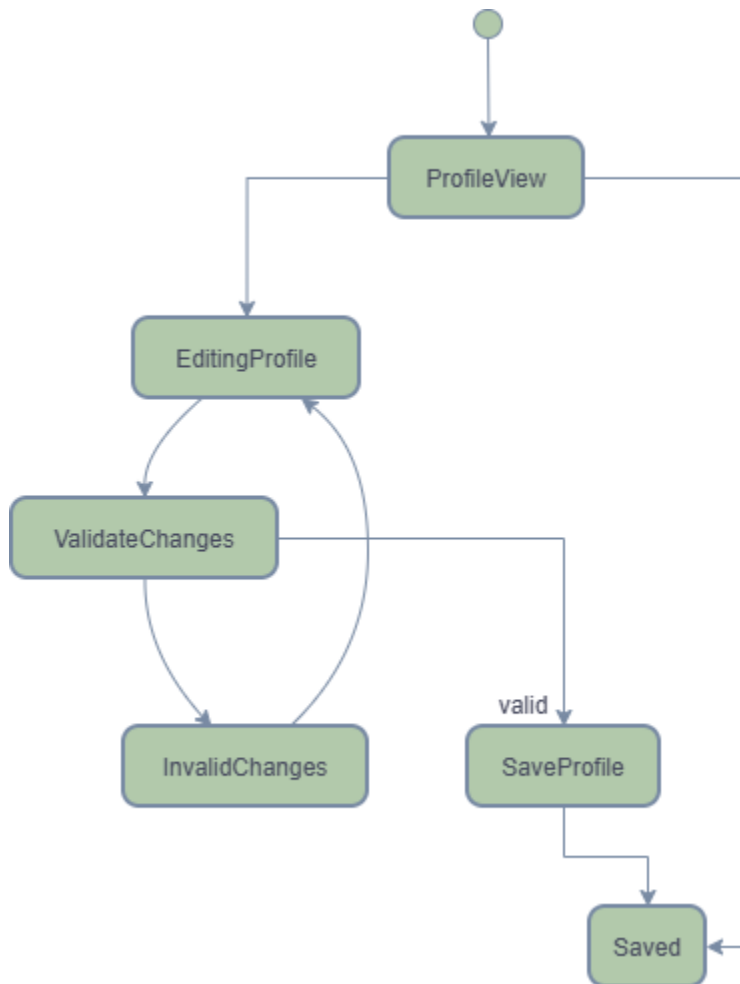
Describes moving through lessons, watching videos, completing modules, and tracking progress.

5.2.2.6 State Diagram 6

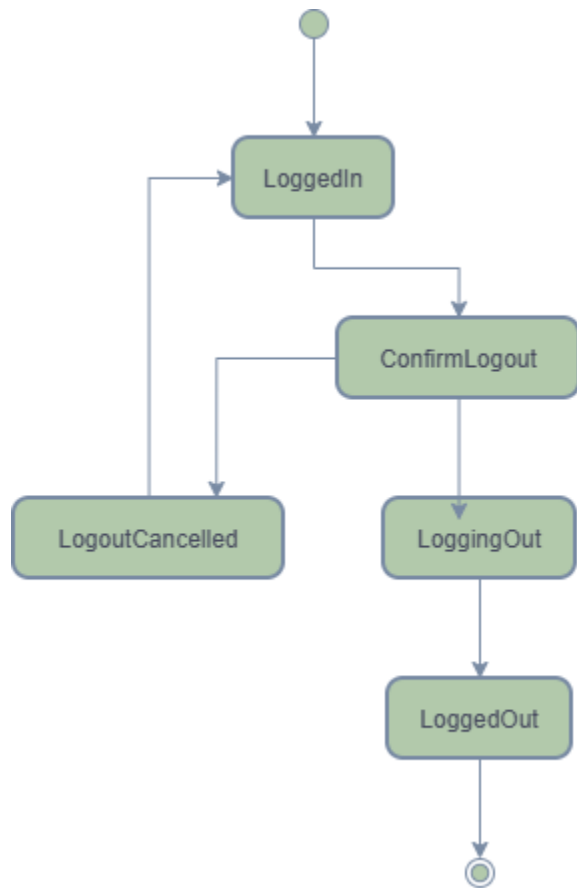


Represents the flow of starting an assessment, answering questions, processing responses, and evaluating the score.

5.2.2.7 State Diagram 7



Covers editing profile, saving updates, handling validation errors.

5.2.2.8 State Diagram 8

Represents the user sign-out flow.

6 References

- Zhou, H., Zhou, W., Qi, W., Pu, J., & Li, H. (2021). Improving Sign Language Translation with Monolingual Data by Sign Back-Translation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2021)*, 1322-1332.
- Liang, Z., Li, H., & Chai, J., "Sign Language Translation: A Survey of Approaches and Techniques," *Electronics*, vol. 12, no. 12, article 2678, 2023, doi:10.3390/electronics12122678.
- Farooq, U., Mohd Rahim, M. S., Sabir, N., Hussain, A., & Abid, A., "Advances in Machine Translation for Sign Language: Approaches, Limitations, and Challenges," *Neural Computing and Applications*, vol. 33, pp. 14357-14399, 2021.

7 Appendices

PSL website by the Family Educational Services Foundation (FESF) used for dataset and lesson plans:

<https://psl.org.pk/>